INCH-POUND
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SUPERSEDING
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MILITARY SPECIFICATION

COMPOSITION B

Inactive for new design after 18 January 2013

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of high explosive designated as Composition B for use in ammunition.

1.2 Classification. Composition B shall be of the following types as specified (see 6.1):

TYPE I – Flake
Grade A – Nominal Viscosity of 5 seconds
Grade B – Nominal Viscosity of 12 seconds

TYPE II – Granular

Comments, suggestions, or questions on this document should be addressed to Commander, US Army ARDEC, RDAR-EIQ-SA, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1376

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-398 – RDX (Cyclotrimethylenetrinitramine)
MIL-T-248 – Trinitrotoluene (TNT)
MIL-W-20553 – Wax, Desensitizing

STANDARDS

MIL-STD-650 – Explosive; Sampling, Inspection and Testing
MIL-STD-1168 – Ammunition Lot Numbering and Ammunition Data Card

DRAWINGS

ORDNANCE CORPS

7548644 – Box, Packing for High Explosives, Assembly
7548645 – Carton, Packing; Reusable-Collapsible for High Explosives, Assembly, Details, Packing and Marking
9257923 – Box, Fiberboard, Packing, Reusable, Collapsible, F/High Explosives, Single Piece, Assembly, Details, Packing and Marking

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E203 – Standard Test Method for Water Using Volumetric Karl Fischer Titration

(Copies of ASTM standards are available online at www.astm.org).

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3. REQUIREMENTS

3.1 Material. RDX complying with the requirements of Specification MIL-R-398, except for granulation, and desensitizing wax complying with the requirements of Specification MIL-W-20553 shall be thoroughly and uniformly incorporated with molten trinitrotoluene (TNT) complying with the requirements of Type 1 of Specification MIL-T-248. These shall be combined in the proportion specified in 3.2 to form a mixture free from gritty particles or other visible impurities.

3.2 Physical and chemical properties. Composition B shall conform to the limits specified in Table I, when determined as specified in the applicable subparagraphs of 4.4.

TABLE I.

Property	Type I	Type II
RDX, %	59.5 ± 2.0	59.5 ± 2.0
TNT, %	39.5 ± 2.3	39.5 ± 2.3
Desensitizer, %	1.0 ± 0.3	1.0 ± 0.3
Moisture, % Max	0.20	0.20
Bulk Density, gm/cc min	-	0.90
Insoluble Particles, max(retained on USS No. 60 screen)	5	5
Viscosity, Efflux sec, max Grade A Grade B	 7.0 17.0	 - -
Granulation, (%) Thru USS No.10 screen, min	-	100
Thru USS No.200 screen, max	-	1.0

3.3 First article testing. This specification makes provision for First Article Testing. Submission of First Article Quantity by the contractor shall be as specified in the contract.

3.4 Workmanship. The material furnished under this specification shall be free of dirt, chips and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the

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contract is submitted to the Government for acceptance , the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in the specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, his authority is vested in another agent of the certifying organization.

4.1.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.2)
2. Quality Conformance Inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a first article sample consisting of 5 lb of Composition B in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government.

Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer (see 6.1).

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

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4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Quality Conformance Inspection.

4.3.1 Lot Formation. A lot shall consist of one or more acceptable batches of Composition B produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions not to exceed 31 consecutive calendar days, unless otherwise stated in the solicitation. Each batch shall consist of that quantity of Composition B that has been subjected to the same unit chemical and physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. Type I TNT from one interfix lot number, from one manufacturer, produced under one set of continuous operating conditions.

b. RDX of one type only, from one manufacturer produced under one set of continuous operating conditions.

c. Wax from one manufacturer.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105). Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and am AQL of 0.65 percent for each Minor defect.

4.3.2.1	Wooden box or fiberboard carton, prior to closing (see Drawing (dwg) 7548644, 7548645, or 9257923 as applicable)		
Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical	None defined.		
Major	AQL 0.65 percent		
101	Liner pierced or torn	Visual	01001
102	Liner improperly closed	Visual	01002
103	Foreign matter	Visual	01003

4.3.2.2	Sealed wooden box (see dwg 7548644)		
Categories	Defects	Method of Inspection	Code No.
Critical	None defined.		
Major	AQL 0.65 percent		
101	Box damaged	Visual	02001
102	Top improperly assembled	Visual	02002
103	Strapping broken, or loose	Visual/Manual	02003

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Minor	AQL 1.50 percent		
201	Nail protruding	Visual	02004
202	Marking misleading or unidentifiable	Visual	02005
203	Strapping improperly assembled	Visual/Manual	02006

4.3.2.3	Sealed fiberboard carton (see dwg 7548645 or 9257923 as applicable)		
Categories	Defects	Method of Inspection	Code No.
Critical	None defined.		
Major	AQL 0.65 percent		
101	Assembly torn or pierced	Visual	03001
102	Lot number misleading or unidentifiable	Visual	03002
103	Strapping broken or loose	Visual/Manual	03003
Minor	AQL 0.65 percent		
201	Marking misleading or unidentifiable	Visual	03004

4.3.3 Testing.

4.3.3.1 Sampling for Type I Composition B (see 3.2). A representative sample of 2 pounds shall be taken from every batch of Composition B produced. A 550 gm sample shall selected from the 2 pound sample and tested for viscosity in accordance with paragraph 4.4.3. The remainder of the 2 pound sample shall be retained for testing of the other characteristics listed in paragraph 3.2 as required by the sampling frequency.

The remainder of the 2 pound samples representing the first eight batches produced in each incorporation building shall be tested for compliance to the specification requirements. If any sample fails to meet the applicable requirements the batch represented by the sample shall be rejected. Samples representing additional batches from the same building which produced the defective material shall be tested for compliance to the specification requirements until eight consecutive batches produced in the building meet all specification requirements. After eight consecutive batches meet the above requirement, the building will be on a sampling frequency for all characteristics except viscosity. Two separate samples from the remainder of the 2 pound samples representing the number of batches produced in an incorporation building during a shift will be randomly selected and tested for compliance to the specification requirements. Failure of a sample to comply with any of the requirements specified herein requires rejection of the batch represented by the sample. At this point, the material produced previously on the same casting belt in the building will be screened for the defect in reverse production sequence until either a batch meets the specification or until a previously accepted batch is reached, whichever occurs first. In addition 100% inspection will be in effect from this point on until eight consecutive

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batches produced in the building meet all specification requirements. Since every batch is sampled but not tested due to testing frequency of 2 batches per shift per incorporation building after eight consecutive batches meet all specification requirements, the samples that remain untested can be returned to their respective batches.

4.3.3.2 Sampling for Type II Composition B (see 3.2). A representative sample of approximately 300 grams shall be selected from every batch of Composition B and tested for conformance to the requirements listed in Table I of paragraph 3.2. Failure of a sample to comply with any of the requirements specified herein requires rejection of the batch represented by the sample.

4.3.3.3 Sample preparation. Examine the representative sample obtained in paragraph 4.3.3.1 visually for foreign material. Then crush and screen the material, and collect that portion of the material passing through a USSS No.4 and retained on a USSS No. 16 for the tests described in paragraph 4.4.1, 4.3.2, and 4.4.4.

4.3.3.4 Testing of Composition B (see 3.2). The samples of Type I and Type II material shall be tested for the applicable properties listed below:

Property	Type of Defect	Code Number	Test Paragraph
RDX content	Major	04001	4.4.1
Desensitizer content	Major	04002	4.4.1
TNT content	Major	04003	4.4.1
Moisture content	Major	05001	4.4.2
Viscosity	Major	06001	4.4.3
Insoluble particles	Major	07001	4.4.4
Granulation	Major	08001	4.4.5
Bulk density	Major	09001	4.4.6

4.3.4 Inspection Equipment. The Government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.3 and 4.4. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of a11 inspection equipment is required prior to its use for acceptance purpose.

4.4 Test methods and procedures.

4.4.1 Determination of ingredients.

4.4.1.1 RDX content. A weighed portion of 10.00 grams (gms) of the sample shall be placed in a 100 milliliter (mL) beaker, and 30 mL of toluene saturated with RDX shall be added. The beaker shall be covered with a watch glass and placed on a steam bath for 30 minutes. The lumps shall be broken up with a glass rod and the solution agitated occasionally by swirling. After cooling to room temperature the solution shall be filtered through a tared filtering crucible.

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The insoluble residue shall be transferred from the beaker to the filtering crucible using four portions or more of 2 to 3 mL of each of toluene saturated with RDX. Air shall be drawn through the crucible until the odor of toluene is no longer detectable. The crucible and contents shall be dried for one hour at 100 degrees C., plus or minus 5 degrees C., cooled in a desiccator, and weighed. The percentage of RDX shall be calculated as follows: (see 6.3)

Percent RDX = 100 A divided by (W – MW)

Where:

A = increase of weight of filtering crucible, gms
W = weights of sample, gms
M = percent moisture in material, expressed as a decimal (see paragraph 4.4.2)

4.4.1.2 Desensitizer content. 5.00 gm of a finely ground portion of the sample shall be weighted into a tared filtering crucible of coarse porosity. A 15 mL portion of a reagent grade acetone previously cooled to 5 degrees C., shall be added, the crucible and contents swirled, and suction applied. This operation shall be carried out in such a manner that the time of contact between sample and acetone approximates 5 seconds. This procedure shall be repeated nine times making a total of ten 15 mL extractions. The sides of the crucible shall be washed down each time to insure removal of the TNT and RDX. The crucible and residual wax shall be aspirated for 5 minutes, allowed to remain in the desiccator for 5 minutes, and weighed. The increase in weight, as percent desensitizer in the sample, shall be calculated as follows:

Percent desensitizer = 100 (A plus 0.0009 (W – MW)) by (W – MW)

Where:

A = gm of residual wax
W = gm of sample
M = percent moisture in material, expressed as a decimal (see paragraph 4.4.2)
0.0009 = compensating factor for desensitizer solubility in acetone

4.4.1.3 TNT content. The TNT content shall be determined by difference. From 100.0 percent, the sum of the percentages of RDX and desensitizer shall be subtracted. The difference is the percent TNT.

4.4.2 Determination of moisture content (see 6.4).

4.4.2.1 Special solvent. The special solvent shall consist of equal volumes of anhydrous methanol and toluene thoroughly mixed. If necessary the solvents shall be dried by distillation.

4.4.2.2 Procedure (Karl Fischer method). Determine the moisture content in accordance with ASTM E203 Standard Test Method for Water Using Volumetric Karl Fischer Titration.

4.4.3 Determination of viscosity (efflux method). The viscosity shall be determined in accordance with Method 212.1 of MIL-STD-650.

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4.4.4 Insoluble particles. Determine the insoluble particles in accordance with Method 106.1 of MIL-STD-650. Sufficient solvent shall be added to completely dissolve a 50 gram sample.

4.4.5 Granulation. Determine the granulation in accordance with Method 204.1 of MIL-STD-650.

4.4.6 Bulk density. Determine the bulk density of the granular Composition B in accordance with Method 201.1 of MIL-STD-650.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Composition B shall be packed and marked in accordance with dwg 7548644.

5.1.2 Level B or level C. Composition B shall be packed and marked in accordance with dwg 7548645 or dwg 9257923.
5.1.3

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:
a. Title, number and date of this specification.
b. Type and grade required (see 1.2)
c. Level of packing protection required.
d. Description sheets shall be prepared for each lot in accordance with MIL-STD-1171.
e. Provisions for submission of first article samples.

6.2 Inspection code numbers. The five-digit code number assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 RDX content. A reduced sample size of 5.00 gms and 15 mL of solvent may be used in place of the quantities specified in 4.4.1.1 to conform to local safety standards.

6.4 Moisture Determination. Analytical Standard Method Number C-3 dated 4 October 1971 by the Holston Defense Corporation can be used in place of the Karl Fisher Method to determine the moisture content. Any other method must be approve d by Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SARPA-QA-A-P prior to use.

6.5 Incorporated changes. ECP R17Y5004 and R11Y5002 provides documentation for noted changes.

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6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - AR
Navy - OS
Air Force - 70

Preparing Activity:
Army- AR
(Project 1376-2019-009)

Review activities:

Navy- SH
Air Force-99

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